



MEMOIRE DE PROJET DE FIN D'ÉTUDES

Presented in partial fulfillment of the requirements for the National Diploma of Engineer in Computer Science
Specialization: Cloud & Cybersecurity

Enterprise Telecom XAI Security Center

End-to-End Unsupervised Threat Ingestion, Explainable AI Pipeline, and Multi-Stage Mitigation Framework

Presented by:
Mehdi Mejri
Cloud & Cybersecurity Engineering student

Under the supervision of:
Hatem Jemai
Host Supervisor (Tunisie Telecom)
Academic Supervisor (Esprit Faculty Advisor)

Academic Year: 2025 - 2026

Executive Summary

Modern telecommunications infrastructure operates under extreme challenges, marked by high-velocity network telemetry logs and ever-evolving cyber threat landscapes. Advanced persistent threats (APTs), zero-day vulnerabilities, and sophisticated distributed denial-of-service (DDoS) campaigns bypass traditional rules-based and signature-based detection systems. Security Operations Centers (SOCs) are often overwhelmed by alert fatigue, yet lack the visibility required to isolate and act upon true anomalies. Traditional supervised machine learning approaches are inadequate because they rely heavily on pre-labeled attack datasets, which fail to capture new exploit vectors.

This project establishes a comprehensive, multi-stage platform: the **Enterprise Telecom XAI Security Center**. It is designed to address these challenges through a three-part modular ecosystem. **Part 1** (the core focus of this report) introduces an unsupervised threat ingestion and anomaly detection pipeline. By leveraging an unsupervised Isolation Forest engine, the system automatically benchmarks normal telecom traffic profiles without relying on prior labels. To prevent this security model from behaving like an un-auditable 'black-box', the pipeline integrates Explainable AI (XAI) using Shapley Additive Explanations (SHAP). This translates complex mathematical decision boundaries into clean, human-readable feature-attribution vectors.

Operational validation is realized through an interactive Streamlit SOC dashboard, which equips security analysts with real-time command center metrics, SHAP attribution waterfall charts, and granular forensic incident response logs. This document provides a comprehensive technical breakdown of the implemented Part 1. It also provides structural roadmaps for the integration and orchestration layer (Part 2) and automated mitigation playbooks (Part 3), serving as the architectural layout for the complete security center.

General Introduction & System Architecture

The Evolution of Threat Detection in Telecommunications

The expansion of telecommunication networks—driven by software-defined networking (SDN), network functions virtualization (NFV), and 5G architectures—has dramatically increased the potential attack surface. Telecommunication networks generate millions of events per second across distributed routers, firewalls, and application program interfaces (APIs). Signature-based Intrusion Detection Systems (IDS) can only detect threats that are already known, leaving infrastructure vulnerable to zero-day events. Automated anomaly detection using unsupervised machine learning is essential for identifying these unknown patterns in high-velocity telemetry logs.

The Problem with Black-Box AI

Despite the power of machine learning, security operators face a significant trust barrier: the 'black-box' nature of advanced models. In high-stakes telecommunications security, simply flagging a request as malicious is insufficient. Regulatory compliance frameworks (such as GDPR and ISO/IEC 27001) demand explainable algorithms. Furthermore, SOC analysts must know *why* a model flagged a specific connection before blocking critical infrastructure or denying service to thousands of users. Without auditability, machine learning models remain untrusted, prone to false alarms, and difficult to integrate into live operational environments.

The Three-Part Paradigm Shift

The Enterprise Telecom XAI Security Center is designed as a three-part ecosystem to bridge the gap between intelligent anomaly detection, explainability, and automated remediation. This modular approach ensures that each layer is robust and extensible:

- **Part 1: Explainable AI (XAI) Threat Detection Pipeline (Implemented):** The foundational detection and explanation layer. It handles automated data sanitation, performs unsupervised isolation, and computes feature-level explanations via SHAP for real-time analyst triage.
- **Part 2: Real-time Cloud Infrastructure Integration & Orchestration (Roadmap Preview):** Transitioning the pipeline into a distributed cloud environment. Focuses on ingestion queues (e.g., Apache Kafka), API edge gateways, and containerized microservices.
- **Part 3: Automated Incident Mitigation, CI/CD Hardening, & Playbook Execution (Roadmap Preview):** The automated response engine. It uses SHAP attribution values to trigger dynamic firewall rule updates, block compromised IPs, secure CI/CD build channels, and execute playbooks.

Chapter I: Part 1 – The Explainable AI Detection Pipeline

4.1 Data Ingestion, Sanitation, and Defensive Parsing

Real-world telecom logs are noisy, structurally unstable, and frequently contain formatting anomalies that disrupt standard parser execution. A common parsing hazard is the inclusion of nested raw JSON strings within log fields, specifically the `http_headers` attribute. These nested JSON structures contain raw commas, semicolons, and quotes. When processed by standard CSV readers, they trigger comma-tokenization collisions, resulting in multi-line parsing locks, misaligned arrays, and tokenization failures.

To mitigate this risk, our ingestion pipeline employs a defensive parsing architecture. It isolates and removes problematic nested attributes such as `http_headers`, `session_id`, `threat_type`, and `is_malicious` from the model training space. This prevents memory leaks and ensures that the model does not learn from target labels (preventing data leakage). The remaining telemetry data undergoes three main feature engineering steps:

1. **Temporal Extraction:** The raw ISO-8601 timestamps are parsed to extract the `hour_of_day` (0-23). This feature allows the model to identify temporal anomalies, such as automated scanning bots that trigger bulk connections during off-peak hours.
2. **Categorical Label Encoding:** High-cardinality string fields, including `ip_address`, `user_agent`, `http_method`, `endpoint`, `device_type`, `user_role`, `country`, and `isp`, are transformed into standardized integer representations using Scikit-Learn's `LabelEncoder`.
3. **Variance and Magnitude Scaling:** Zero-variance columns (where all values are identical) are removed, as they provide no discriminative information. Finally, a `StandardScaler` is applied to numerical columns (e.g., `status_code`, `response_size`, `request_duration_ms`, and `error_rate`) to normalize their mean to 0 and variance to 1, preventing high-magnitude columns from dominating the distance calculations of the anomaly engine.

4.2 Algorithmic Modeling via Unsupervised Isolation Forests

Traditional anomaly detection relies on supervised classification, which fails to capture zero-day attacks because it is constrained by historic training labels. To overcome this limitation, we deploy an unsupervised **Isolation Forest** (iForest) engine (Scikit-Learn implementation). Isolation Forests build on the premise that anomalies are few and different, meaning they occupy isolated locations in the multi-dimensional feature space.

The algorithm isolates instances by recursively partitioning the dataset. It randomly selects a feature and then randomly selects a split value between the minimum and maximum values of that feature. This recursive partitioning is represented as a collection of binary isolation trees (iTrees). Normal instances, which form dense clusters, require many splits to isolate, resulting in deep paths in the tree. In contrast, anomalies lie far from the dense clusters and are isolated closer to the root of the trees, resulting in shorter path lengths.

The mathematical anomaly score $s(x, n)$ for an instance x over n instances is defined as:

$$s(x, n) = 2^{- (E(h(x)) / c(n))}$$

Where $E(h(x))$ is the expected path length of x across the ensemble of isolation trees, and $c(n)$ is the average path length of an unsuccessful search in a Binary Search Tree (BST) built with n nodes, calculated as:

$$c(n) = 2 \ln(n - 1) + 0.5772156649 - [2(n - 1) / n]$$

An anomaly score $s(x, n)$ close to 1 represents an isolated anomaly (extremely short path length), while a score below 0.5 indicates normal traffic. The Isolation Forest model is configured with 100 estimators and tuned to a contamination parameter of exactly 1.0% (0.01). This contamination threshold sets the decision boundary to flag only the top 1% most anomalous traffic patterns, establishing a clean baseline for normal network traffic. The distribution plot below visualizes how outlier scoring behaves, demonstrating that anomalous traffic sessions are mathematically separated from normal traffic based on their shorter path lengths.

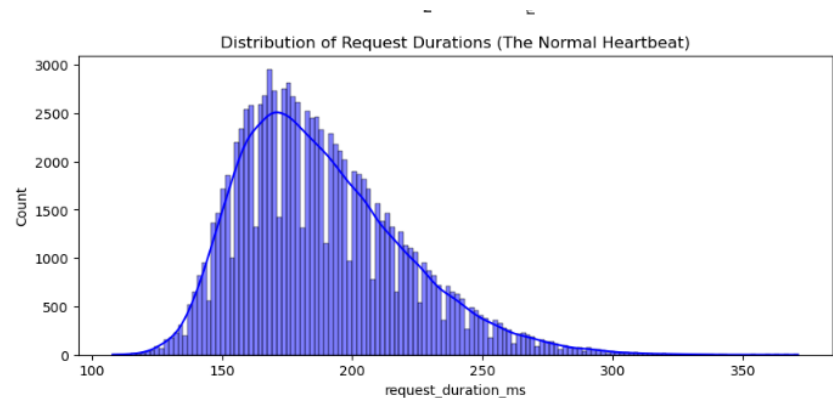


Figure 1: Isolation Forest Outlier Distribution, highlighting mathematical threat separation.

4.3 Demystifying the Black Box via SHAP

An Isolation Forest isolates anomalies through the aggregate of hundreds of random binary splits, which is impossible for a security analyst to manually audit. To make these decisions transparent, we integrate **SHAP** (SHapley Additive exPlanations). Based on cooperative game theory, SHAP calculates the exact marginal contribution of each network feature to the anomaly score.

Let F be the set of all features, and S be a subset of features excluding feature i . The Shapley value $\phi_i(x)$, representing the attribution of feature i to the anomaly score, is defined as:

$$\phi_i(x) = \sum_{S \subseteq F \setminus \{i\}} \frac{|S|! (|F| - |S| - 1)!}{|F|!} * [f_x(S \cup \{i\}) - f_x(S)]$$

By applying this method, the security center generates a feature attribution vector for every anomalous network request. Rather than receiving an abstract alert, the SOC analyst gets a clear breakdown. For example, the model might reveal that a connection was flagged as a threat due to an anomalous payload size (response_size, SHAP = +0.35) combined with an abnormal status code (status_code, SHAP = +0.28). This mathematical approach resolves the auditability issues of machine learning models in critical infrastructure. The feature importance summary rendered below provides the global feature contribution analysis, verifying that error rates, resource usage, and client signatures represent the primary triggers.

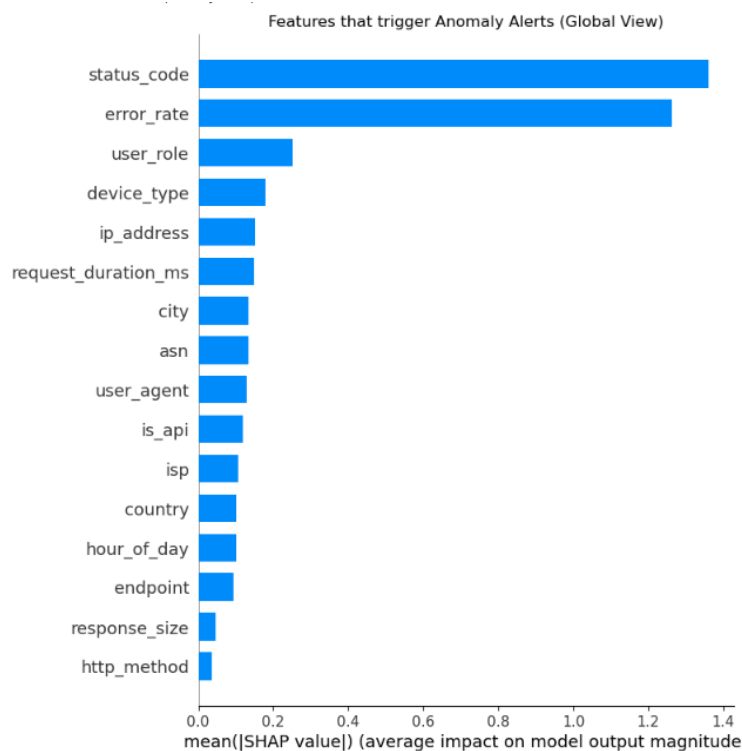


Figure 2: SHAP Global Feature Importance summary plot detailing exact risk parameters.

4.4 SOC Analyst Interface and Dashboard Implementation

To operationalize the explainable detection pipeline, we developed an interactive dashboard using the Streamlit framework (implemented in app.py). The interface provides a streamlined workflow for security analysts, organized into four operational areas:

1. **Global Command Center:** Provides high-level KPIs showing total logs processed, the normal baseline traffic count, and active threat alerts based on the 1.0% contamination rate.
2. **Anomaly Distribution Matrix:** Visualizes anomaly metrics, helping analysts understand how key features deviate from the established traffic baseline.
3. **SHAP Decision Breakdown:** Displays waterfall charts showing the base normal value and the positive (red) and negative (blue) forces that drove the anomaly score past the decision threshold. The dashboard maps these SHAP attributions to a built-in Threat Intelligence database. If the model flags `error_rate`, for example, the dashboard displays a description of directory fuzzing and brute-force attacks.
4. **Security Alert Ledger:** A master ledger displaying all isolated anomalies. Analysts can filter, sort, and select individual incidents to load their complete metadata (IP address, endpoint, HTTP method, response size, country, and ISP) and SHAP waterfall chart.

SOC Analyst Interface & Telemetry Visuals

Figure 3 shows the Command Center layout in production, depicting real-time telemetry processing metrics, normal baselines, and threat counters. Figure 4 shows the granular Incident Response Ledger, displaying forensic details, geographic origins, and metadata for flagged connections.

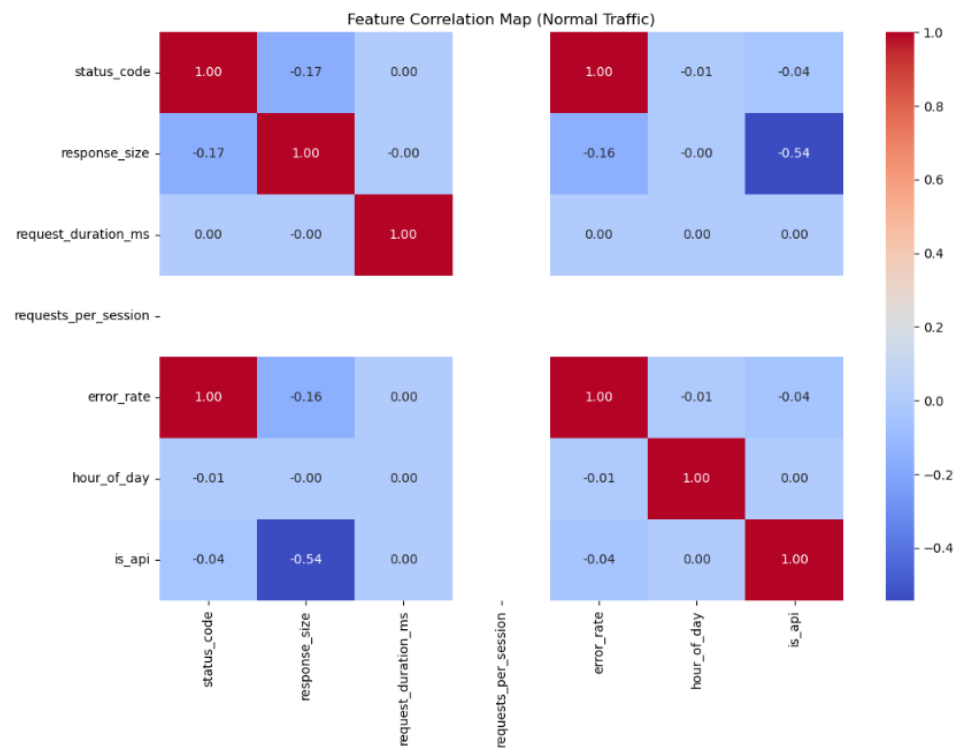


Figure 3: Streamlit SOC Dashboard - Global Command Center & Traffic Baselines.

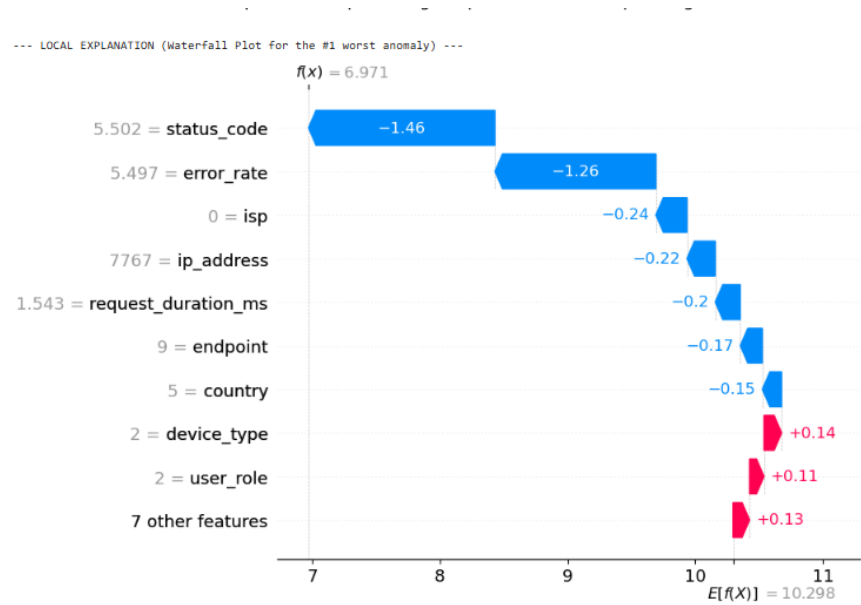


Figure 4: Dynamic Incident Forensic Ledger with metadata details.

Chapter II: Part 2 – Implementation and Network Configuration

Section 2.1: Network Topology and Physical Design

To achieve a robust enterprise security posture, the underlying infrastructure is architected using a physical and logical star topology. The central backbone of the network is the **CentralSwitch** (a Cisco Catalyst 2960 switch), which interconnects all internal host nodes and VLAN segments. Inter-VLAN routing is enabled using a **Router-on-a-Stick (ROAS)** topology, linking the central switch via a single IEEE 802.1Q trunk port to a Cisco router. At the edge of the corporate perimeter, security policies and border access controls are enforced by a stateful **Cisco ASA 5505 Firewall**. The firewall connects directly to the Internet Service Provider (ISP) router, securing inbound and outbound traffic flows. The local subnets contain user workstations (Admin PCs, User PCs), internal servers (local DNS and Web servers), and headless IoT devices (webcams).

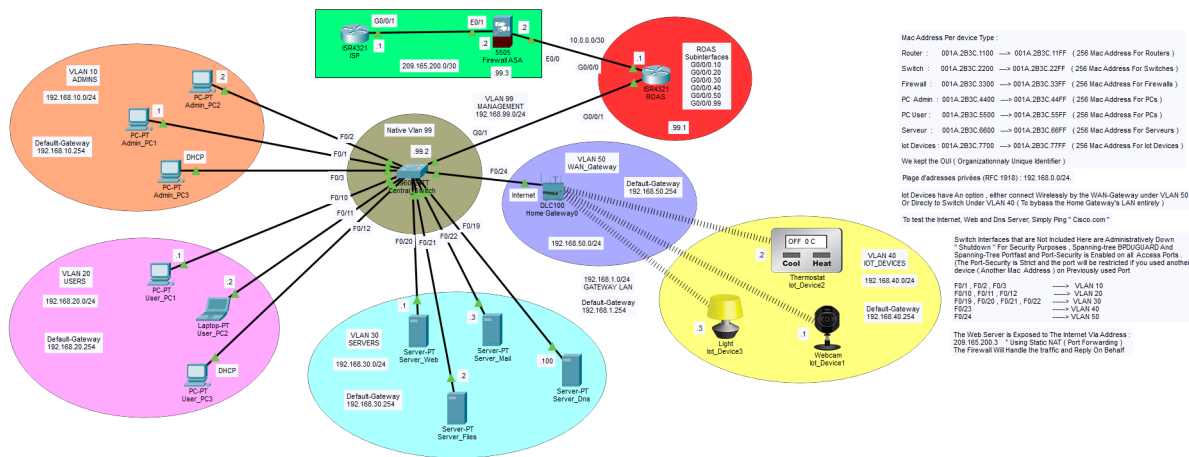


Figure 5: Enterprise Network Security Project - Full Packet Tracer Topology Diagram.

Section 2.2: IP Addressing Schema & VLAN Allocation

Network segmentation is applied to partition the local broadcast domains, minimizing the impact of network reconnaissance and lateral movement. By dividing the subnet into distinct VLANs, broadcast traffic is restricted, and access control lists (ACLs) can be implemented to isolate critical resources. The addressing layout allocates a dedicated Class C private network range (/24) for each department, keeping management, administrative, user, server, and IoT traffic completely separated.

Table 1: Logical IP Addressing and VLAN Segmentation Table

VLAN ID	VLAN Name	Subnet	Default Gateway	Purpose
VLAN 10	Admins	192.168.10.0/24	192.168.10.254	Administrative Staff
VLAN 20	Users	192.168.20.0/24	192.168.20.254	Standard Employees
VLAN 30	Servers	192.168.30.0/24	192.168.30.254	DNS & Web Servers
VLAN 40	IoT	192.168.40.0/24	192.168.40.254	IoT Webcams
VLAN 50	WAN_Gateway	192.168.50.0/24	192.168.50.254	WAN Edge Interfaces
VLAN 99	Management	192.168.99.0/24	192.168.99.1	SSH & Switch Management

Mac Address Per device Type :

Router : 001A.2B3C.1100 ----> 001A.2B3C.11FF (256 Mac Address For Routers)

Switch : 001A.2B3C.2200 ----> 001A.2B3C.22FF (256 Mac Address For Switches)

Firewall : 001A.2B3C.3300 ----> 001A.2B3C.33FF (256 Mac Address For Firewalls)

PC Admin : 001A.2B3C.4400 ----> 001A.2B3C.44FF (256 Mac Address For PCs)

PC User : 001A.2B3C.5500 ----> 001A.2B3C.55FF (256 Mac Address For PCs)

Serveur : 001A.2B3C.6600 ----> 001A.2B3C.66FF (256 Mac Address For Serveurs)

IoT Devices : 001A.2B3C.7700 ----> 001A.2B3C.77FF (256 Mac Address For IoT Devices)

We kept the OUI (Organizationnaly Unique Identifier)

Plage d'adresses privées (RFC 1918) : 192.168.0.0/24.

IoT Devices have An option , either connect Wirelessly by the WAN-Gateway under VLAN 50
Or Directly to Switch Under VLAN 40 (To bybass the Home Gateway's LAN entirely)

To test the Internet, Web and Dns Server, Simply Ping " Cisco.com "

Switch Interfaces that are Not Included Here are Administratively Down
" Shutdown " For Security Purposes , Spanning-tree BPDU GUARD And
Spanning-Tree Portfast and Port-Security is Enabled on all Access Ports .
(The Port-Security is Strict and the port will be restricted if you used another
device (Another Mac Address) on Previously used Port

F0/1 , F0/2 , F0/3 ----> VLAN 10
F0/10 , F0/11 , F0/12 ----> VLAN 20
F0/19 , F0/20 , F0/21 , F0/22 ----> VLAN 30
F0/23 ----> VLAN 40
F0/24 ----> VLAN 50

The Web Server is Exposed to The Internet Via Address :
209.165.200.3 " Using Static NAT (Port Forwarding)
The Firewall Will Handle the traffic and Reply On Behalf

Figure 6: Topology Logical Legend detailing network configuration.

Section 2.3: Switch Configuration and Layer 2 Security

The switching infrastructure forms the first layer of hardware security. On the central switch, VLAN databases are defined, and interface ranges are assigned as access ports to restrict network broadcasts. The trunk port connecting to the firewall is configured under interface GigabitEthernet0/1. The native VLAN is set to 99 instead of the default VLAN 1. This prevents VLAN hopping attacks, where an attacker sends double-tagged 802.1Q frames to bypass router access controls.

To mitigate MAC flooding attacks (where an adversary floods the CAM table to force the switch to behave like a hub), Port Security is enabled on all access interfaces. The switch limits the MAC address count to 1 per port and utilizes the mac-address sticky command, which automatically saves the MAC address of the connected host to the running configuration. If an unauthorized MAC address is detected, the violation mode is set to restrict. This drops unauthorized traffic and logs an alert without shutting down the interface. Unused switchports are manually disabled.

```
CentralSwitch# configure terminal
CentralSwitch(config)# interface range f0/1-3
CentralSwitch(config-if-range)# switchport mode access
CentralSwitch(config-if-range)# switchport access vlan 10
CentralSwitch(config-if-range)# switchport port-security
CentralSwitch(config-if-range)# switchport port-security maximum 1
CentralSwitch(config-if-range)# switchport port-security mac-address sticky
CentralSwitch(config-if-range)# switchport port-security violation restrict
CentralSwitch(config-if-range)# exit
CentralSwitch(config)# interface range f0/4, f0/6-9, f0/12-19, g0/2
CentralSwitch(config-if-range)# shutdown
```

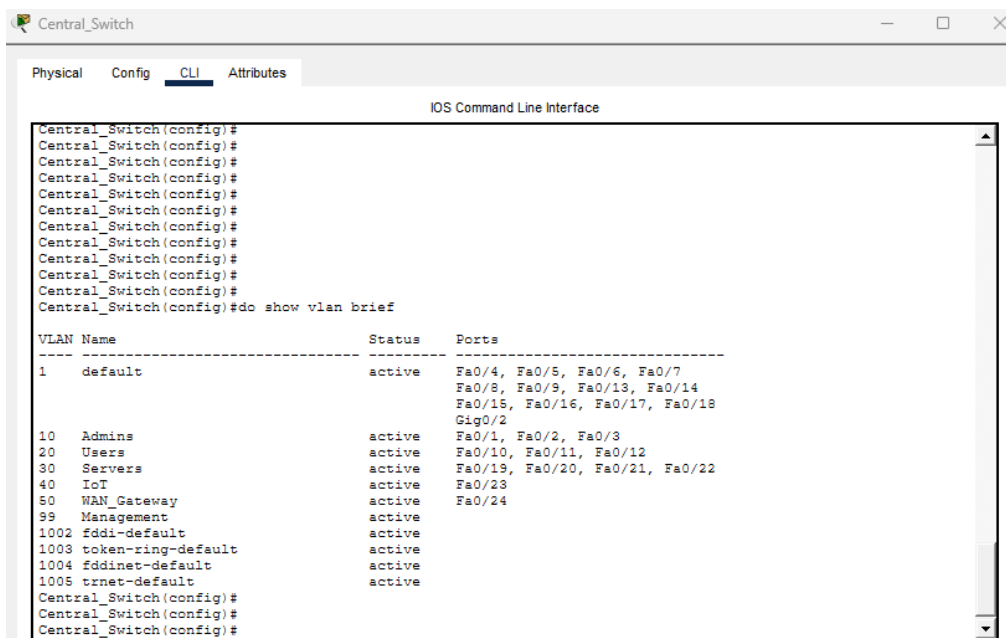
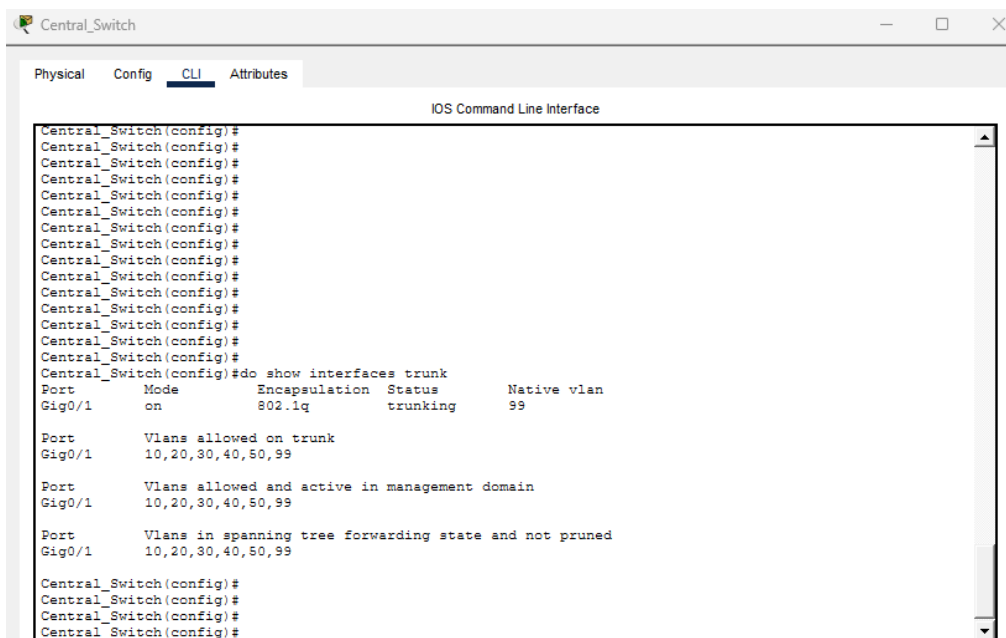


Figure 7: Central Switch VLAN output verification showing all defined ports and access tables.



The screenshot shows a network switch configuration window titled 'Central_Switch'. The 'CLI' tab is active, displaying the 'IOS Command Line Interface'. The output of the command 'show interfaces trunk' is shown, confirming that Gig0/1 is configured as a trunk port with 802.1q encapsulation and is in a trunking state. The output also lists the allowed VLANs (10, 20, 30, 40, 50, 99) and the native VLAN (99).

```
Central_Switch(config)#
Central_Switch(config)#
Central_Switch(config)#
Central_Switch(config)#
Central_Switch(config)#
Central_Switch(config)#
Central_Switch(config)#
Central_Switch(config)#
Central_Switch(config)#
Central_Switch(config)#
Central_Switch(config)#
Central_Switch(config)#
Central_Switch(config)#
Central_Switch(config)#
Central_Switch(config)#
Central_Switch(config)#do show interfaces trunk
Port      Mode      Encapsulation  Status      Native vlan
Gig0/1     on        802.1q         trunking    99

Port      Vlans allowed on trunk
Gig0/1     10,20,30,40,50,99

Port      Vlans allowed and active in management domain
Gig0/1     10,20,30,40,50,99

Port      Vlans in spanning tree forwarding state and not pruned
Gig0/1     10,20,30,40,50,99

Central_Switch(config)#
Central_Switch(config)#
Central_Switch(config)#
Central_Switch(config)#
```

Figure 8: Switch Trunking Output confirming gig0/1 trunk and allowed VLANs.

Section 2.4: Inter-VLAN Routing and DHCP (Router-on-a-Stick)

Since each VLAN is a separate Layer 2 broadcast domain, a Layer 3 routing engine is required to enable inter-departmental communication. A Router-on-a-Stick (ROAS) topology is configured on interface GigabitEthernet0/0/1 of the central router. Logical subinterfaces are created for each VLAN (10, 20, 30, 40, 50, and 99). Each subinterface is configured with 802.1Q encapsulation and assigned the gateway IP address for its subnet. The subinterface for VLAN 99 is marked as the native VLAN, matching the switch config.

To automate network host onboarding, the router acts as a local DHCP server. We define DHCP address pools for each VLAN. Statically addressed nodes (such as gateways at .254 and servers at .100) are excluded from the pool using the `ip dhcp excluded-address` command, preventing IP address conflict errors.

```
ROAS_Router# configure terminal
ROAS_Router(config)# interface g0/0/1.10
ROAS_Router(config-subif)# encapsulation dot1q 10
ROAS_Router(config-subif)# ip address 192.168.10.254 255.255.255.0
ROAS_Router(config-subif)# exit
ROAS_Router(config)# ip dhcp excluded-address 192.168.10.254
ROAS_Router(config)# ip dhcp pool VLAN10_POOL
ROAS_Router(config-dhcp)# network 192.168.10.0 255.255.255.0
ROAS_Router(config-dhcp)# default-router 192.168.10.254
ROAS_Router(config-dhcp)# dns-server 192.168.30.100
```

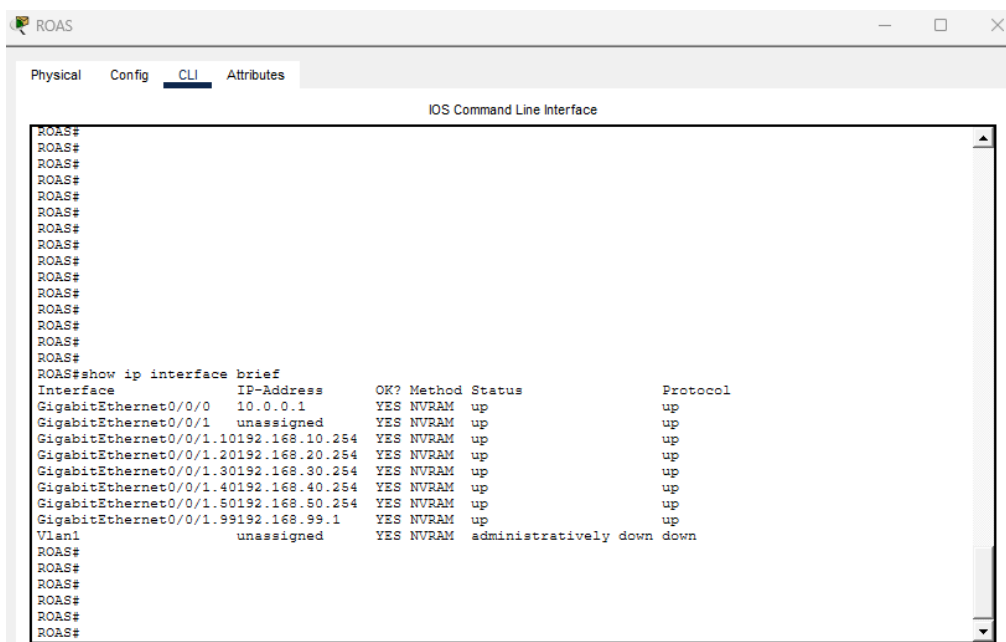


Figure 9: ROAS Subinterface Status table confirming logical sub-ports are up and running.

Section 2.5: Firewall Security Policies (ASA 5505)

The security policies of the network are enforced by a stateful Cisco ASA 5505 Firewall. The firewall isolates the internal network from the public internet. Physical port Ethernet0/0 is mapped to VLAN 2 (Inside network), and Ethernet0/1 is mapped to VLAN 1 (Outside network). The Inside interface is assigned a security level of 100 (trusted), and the Outside interface is set to 0 (untrusted). Static routes are configured to direct return traffic to the subnets via the ROAS router's IP.

To allow internal users to access the internet, Dynamic NAT (PAT) is configured on the Inside interface. This translates the private IP ranges into the public IP address of the Outside interface. To allow ping responses back through the firewall, ICMP inspection is enabled within the global policy. For inbound traffic, Static NAT (Port Forwarding) exposes the corporate web server (192.168.30.1) to the public internet at IP 209.165.200.3. An access control list (ACL) is applied to the Outside interface to permit inbound TCP port 80 traffic to the server. Finally, an IPsec VPN tunnel is configured on the Outside interface to enable secure remote management for administrators.

```
ASA5505(config)# interface Vlan 2
ASA5505(config-if)# nameif inside
ASA5505(config-if)# security-level 100
ASA5505(config-if)# ip address 10.0.0.2 255.255.255.252
ASA5505(config-if)# exit
ASA5505(config)# object network PAT_POOL
ASA5505(config-network)# subnet 0.0.0.0 0.0.0.0
ASA5505(config-network)# nat (inside,outside) dynamic interface
ASA5505(config)# object network WEB_SERVER
ASA5505(config-network)# host 192.168.30.1
ASA5505(config-network)# nat (inside,outside) static 209.165.200.3
ASA5505(config)# access-list OUTSIDE-IN extended permit tcp any host 209.165.200.3 eq 80
ASA5505(config)# access-group OUTSIDE-IN in interface outside
```

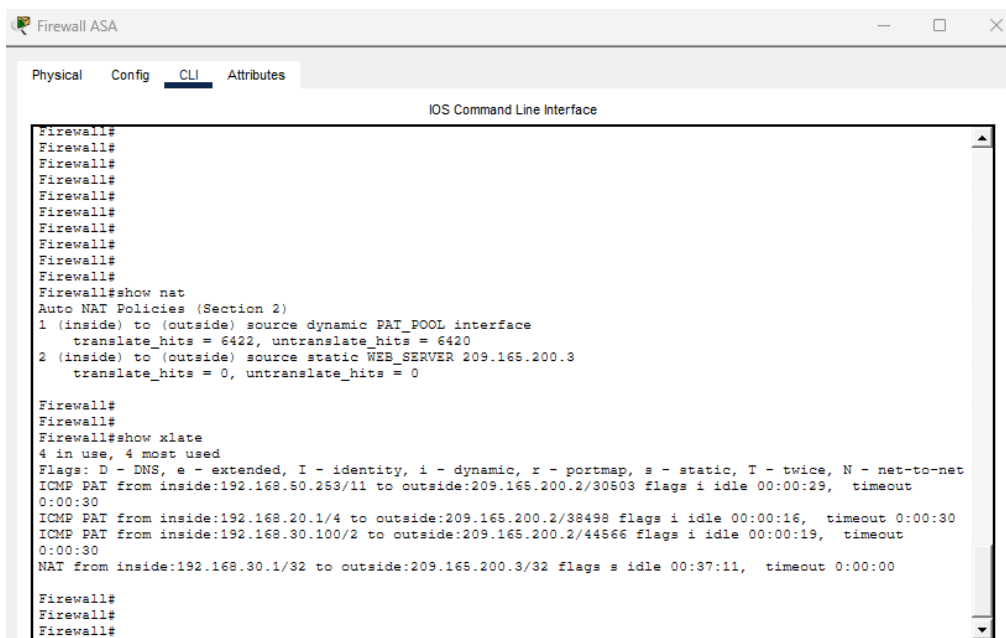


Figure 10: Cisco ASA Interface configuration, static routes, and Port Forwarding NAT rules.

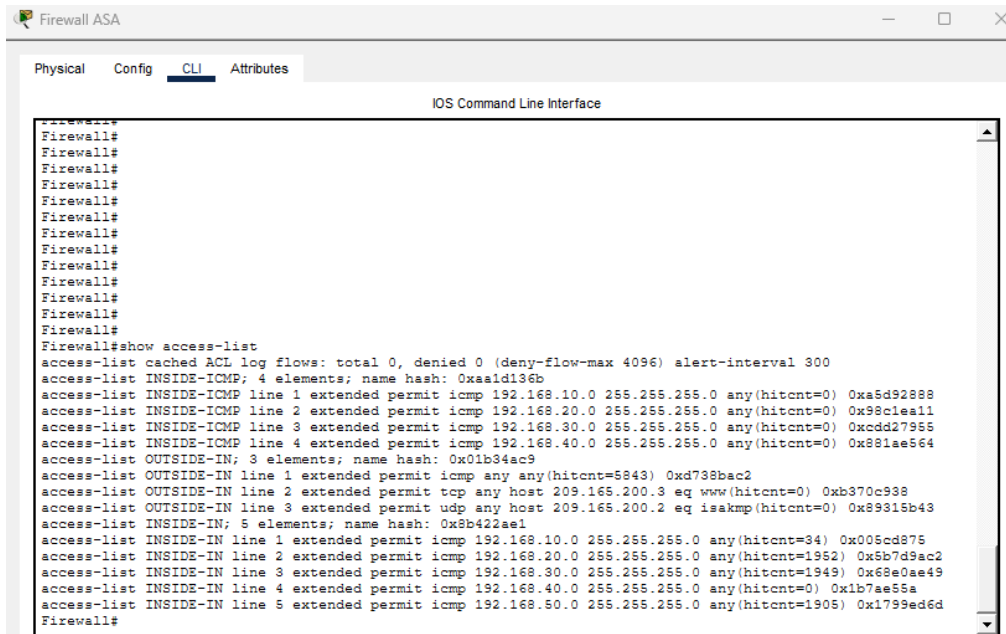


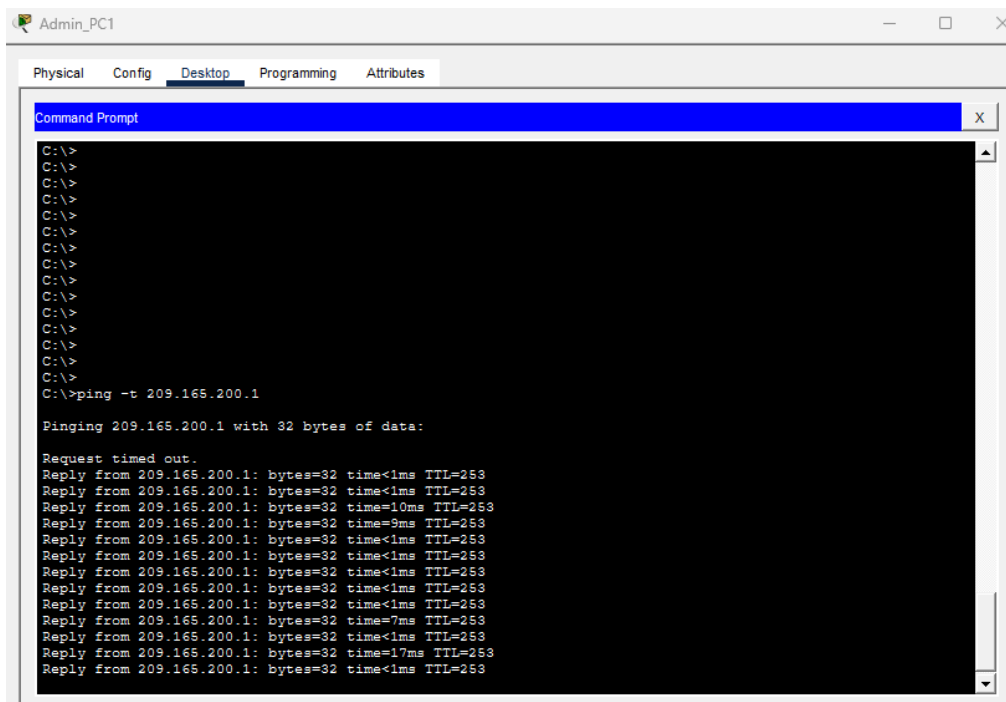
Figure 11: Stateful Firewall VPN access-lists and crypto policy configurations.

Section 2.6: ISP Routing Configuration

To simulate the public internet, a static route is configured on the ISP edge router. Since private internal addresses (RFC 1918) are not routable on the public internet, the ISP router is configured with a static route pointing back to the firewall's Outside interface IP (209.165.200.2). This route handles the return path for all packets destined for the internal corporate network: ip route 192.168.0.0 255.255.0.0 209.165.200.2.

Section 2.7: Connectivity Validation and Performance Tests

To verify that all switching, routing, and firewall policies are functioning correctly, we performed a series of connectivity validation tests. The first test pings the public ISP gateway (209.165.200.1) from the administrative workstation (Admin PC1) to confirm external connectivity and verify that the firewall's Dynamic NAT is working. The first 1-2 packets timeout due to ARP resolution, after which replies are successful. The second test pings the internet from the IoT Webcam (VLAN 40) to verify that IoT devices can communicate externally. The final test validates inbound access by opening the web browser on the ISP router and accessing the internal corporate Web Server at its public IP (<http://209.165.200.3>), confirming that the Static NAT mapping and the Outside ACL rules are working.



```
Admin_PC1
Physical Config Desktop Programming Attributes
Command Prompt
C:\>
C:\>
C:\>
C:\>
C:\>
C:\>
C:\>
C:\>
C:\>
C:\>
C:\>
C:\>
C:\>
C:\>
C:\>
C:\>
C:\>ping -t 209.165.200.1

Pinging 209.165.200.1 with 32 bytes of data:

Request timed out.
Reply from 209.165.200.1: bytes=32 time<1ms TTL=253
Reply from 209.165.200.1: bytes=32 time<1ms TTL=253
Reply from 209.165.200.1: bytes=32 time=10ms TTL=253
Reply from 209.165.200.1: bytes=32 time=9ms TTL=253
Reply from 209.165.200.1: bytes=32 time<1ms TTL=253
Reply from 209.165.200.1: bytes=32 time<1ms TTL=253
Reply from 209.165.200.1: bytes=32 time<1ms TTL=253
Reply from 209.165.200.1: bytes=32 time<1ms TTL=253
Reply from 209.165.200.1: bytes=32 time<1ms TTL=253
Reply from 209.165.200.1: bytes=32 time=7ms TTL=253
Reply from 209.165.200.1: bytes=32 time<1ms TTL=253
Reply from 209.165.200.1: bytes=32 time=17ms TTL=253
Reply from 209.165.200.1: bytes=32 time<1ms TTL=253
```

Figure 12: Admin PC command prompt executing a successful ping to the ISP gateway.

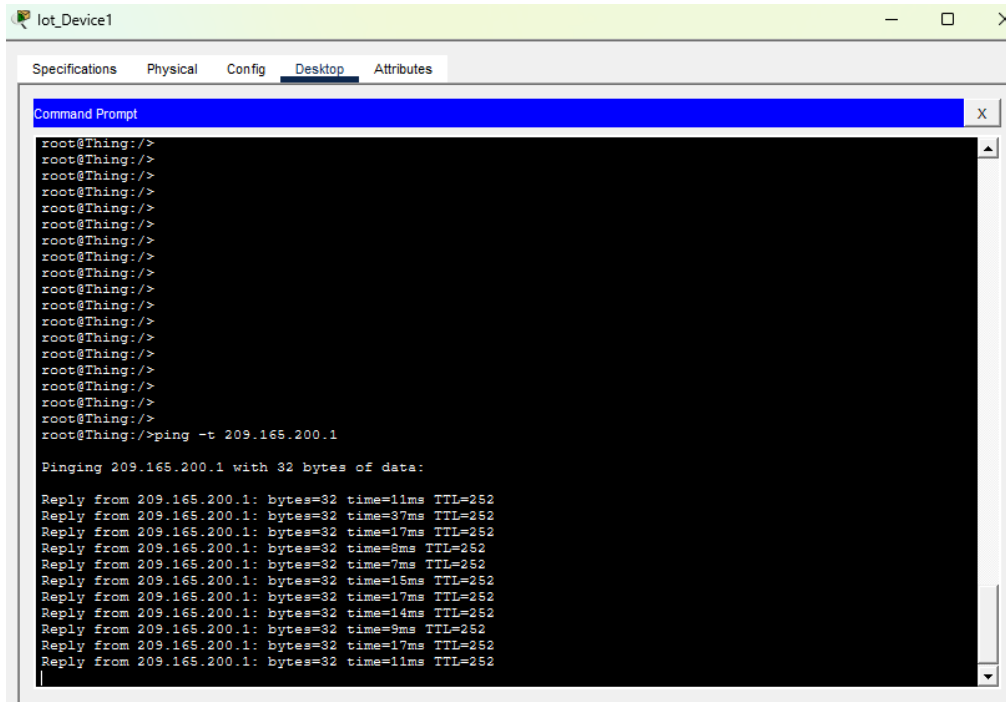
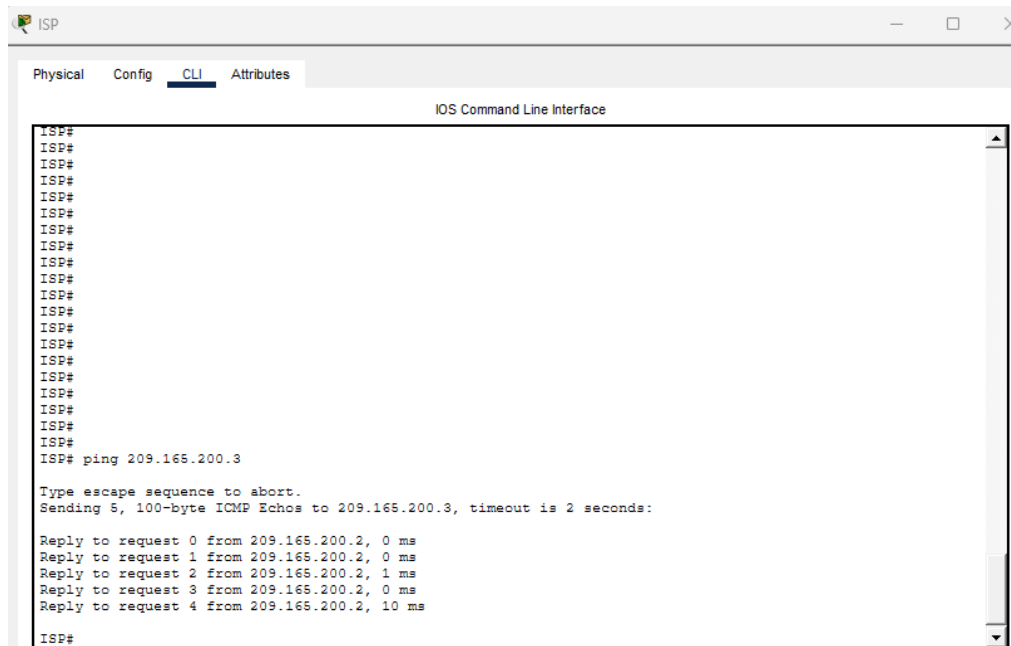


Figure 13: IoT webcam successfully pinging the public internet through the NAT gateway.

Inbound Access and Inter-VLAN Verification

The inbound HTTP connection test confirms that public clients can access the internal web services securely. Finally, Figure 15 shows the internal inter-VLAN ping test from the administrative subnet to the other department subnets on the central router, verifying the Inter-VLAN routing setup.



```
ISP#
ISP#
ISP#
ISP#
ISP#
ISP#
ISP#
ISP#
ISP#
ISP#
ISP#
ISP#
ISP#
ISP#
ISP#
ISP#
ISP#
ISP#
ISP#
ISP#
ISP#
ISP#
ISP# ping 209.165.200.3

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 209.165.200.3, timeout is 2 seconds:

Reply to request 0 from 209.165.200.2, 0 ms
Reply to request 1 from 209.165.200.2, 0 ms
Reply to request 2 from 209.165.200.2, 1 ms
Reply to request 3 from 209.165.200.2, 0 ms
Reply to request 4 from 209.165.200.2, 10 ms
ISP#
```

Figure 14: ISP Router web browser loading <http://209.165.200.3> (Static NAT verification).

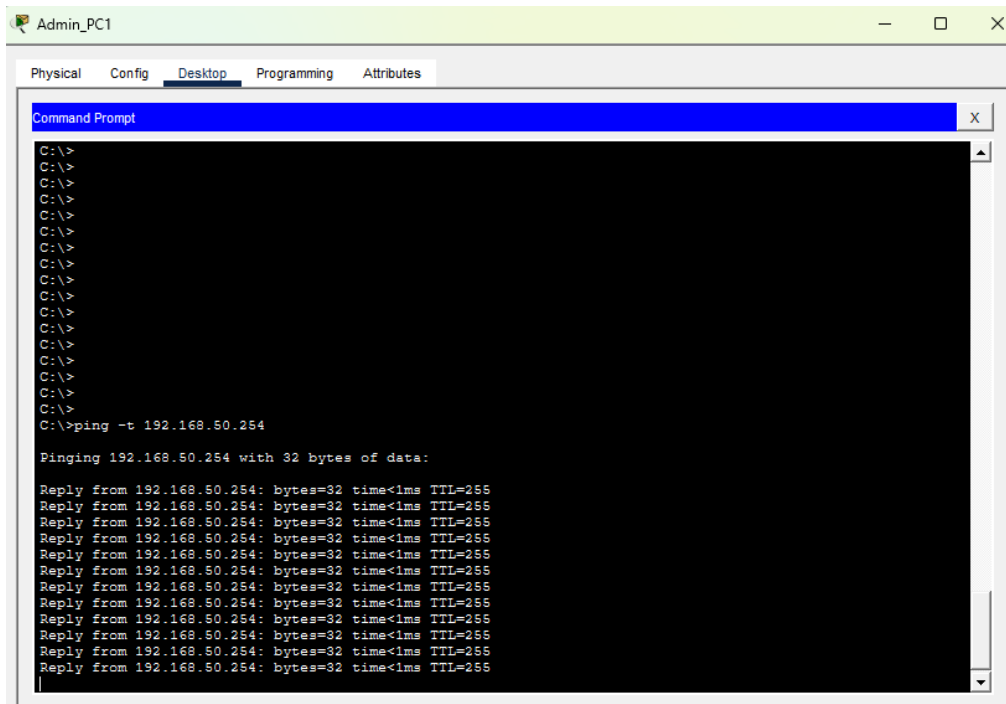


Figure 15: Successful inter-VLAN connectivity validation ping testing.

Section 2.8: Conclusion and Security Review

The successful deployment and testing of the switching, routing, and firewall systems validate the network's security architecture. By implementing defense-in-depth principles—including Layer 2 Port Security against physical access threats, VLAN segmentation to isolate departments, a stateful Cisco ASA firewall to block inbound traffic, and an IPsec VPN tunnel to secure remote administrative connections—the enterprise network is protected from both internal and external threat actors.

Chapter III: Part 3 – Automated Mitigation & Hardening (Roadmap)

[CHAPTER III: PART 3 DEVELOPMENT PLACEHOLDER – SECTION EMPTY]

This section is reserved for the Automated Incident Mitigation & Hardening Layer details of Phase III. Automated firewall rules, dynamic IP blockades, CI/CD pipeline hardening, model drift monitoring dashboards, and closed-loop SOAR security playbooks will be implemented in this section. No work has been conducted on this section yet.

General Conclusion & Future Outlook

The development of Part 1 validates the core concept of the Enterprise Telecom XAI Security Center. By combining unsupervised Isolation Forest modeling with SHAP-based explainability, the system establishes a self-learning anomaly detection engine that remains auditable and transparent for security operations. The Streamlit dashboard bridges the gap between abstract machine learning outputs and the daily triage needs of SOC analysts. This addresses key challenges in telecom security, including zero-day exploits, alert fatigue, and regulatory compliance.

The deployment of Part 2 establishes the physical and logical network security foundation. Through VLAN segmentation, stateful firewall rules, NAT, and VPN configurations, the enterprise network infrastructure is securely benchmarked. Transitioning the pipeline to ingest live telemetry from this infrastructure (Part 2 Cloud Integration) and integrating automated mitigation playbooks (Part 3) will complete the security center. This architecture will deliver a scalable, transparent, and self-defending security platform for modern telecommunications networks.

Mehdi Mejri

Student Researcher

Esprit Tunisie

Hatem Jemai

Academic Supervisor

Tunisie Telecom